Tanmay Bhuskute

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San Francisco, CA

SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with production experience building scalable backend automation, full-stack apps, and quality gates with Java, Python, JavaScript, SQL, and AWS. Researching LLMs for Android testing.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Java, Python, JavaScript, TypeScript, C/C++, SQL, HTML/CSS, Go.
Web & Backend: React, Node.js, Django, Flask, FastAPI, REST APIs, Object-Oriented Design.
Databases & Testing: PostgreSQL, MySQL, DynamoDB, JUnit, Pytest, Jest, Unit Testing, Integration Testing.
Developer Tools: GitLab CI/CD, AWS, Docker, Kubernetes, ArgoCD, Helm, Terraform, SonarQube, CloudWatch, Datadog.
AI & Data: Gemini, Vertex AI, LangChain, MCP, PyTorch, TensorFlow, scikit-learn, pandas, NumPy.

WORK EXPERIENCE

Graduate Researcher | University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline decoupling video understanding from device automation for Android GUI bug reproduction, evaluated against **ViBR** across 53 runs on 20+ apps.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity.
- Built uiautomator2/ADB automation harness with stall detection and config-driven, reproducible evaluation runs.

Software Development Engineer | Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** workflows with Python (Boto3), replacing legacy Ruby scripts and cutting deployment time by **40%**.
- Served as Quality Coach and SonarQube Specialist, increasing automated code coverage from **35% to 78%** via CI/CD quality gates; mentored **5 software engineers** through structured onboarding.
- Executed **44 production environment upgrades** in 4 months, diagnosing live issues across Linux, EC2, databases, and services using CloudWatch metrics and Datadog log correlation.
- Designed **Ansible-based Blue-Green deployment** framework reducing production upgrade downtime by **50%**, with automated rollback restoring full environments within 15 minutes.
- Migrated **40+ production environments** from process-based deployments to Kubernetes using ArgoCD, Helm, GitLab CI/CD, Amazon EFS, and HashiCorp Vault.

Software Developer Intern | Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

Software Developer Intern | Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built full-stack **React and MySQL** application digitizing departmental records and deployed on-premises MySQL with authentication and permissions, cutting document retrieval time by **90%** (5 min to 30 sec).

PROJECTS AND RESEARCH

PolicyPilot | Medical Benefit Policy Intelligence Portal [React, TS, Node, Gemini, MCP] May 2026

- Architected evidence-grounded **RAG pipeline** over Google Gemini 2.5-flash, forcing citation-backed answers to eliminate hallucination in high-stakes healthcare eligibility decisions.
- Designed **17-tool MCP server** exposing structured LLM tool-calling for PA extraction, patient-readiness evaluation, and cross-payer comparison, consumable by external AI agents.

Spendora | AI Personal Finance Platform [Next.js, TS, PostgreSQL, Gemini, Vitest] Jun 2025

- Built Gemini-powered expense entry from text, voice, and receipts, cutting median entry time by **72%** (29s to 8s).
- Implemented Supabase auth and row-level security, with **80 tests** (29 unit, 40 database, 11 end-to-end).

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).